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INTERVIEW

Top ReactJS Interview Questions and Answers Of 2024! [Part-1]

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Do you know that more than 1,300 developers and over 94,000 sites utilize ReactJS? React or React.js or ReactJS is a free and open-source front-end JavaScript library.

If you wish to create modern websites with super performance and the highest security, then React should be your top pick. Top entrepreneurs and developers prefer ReactJS as it allows enterprises



to craft apps with the best user experiences. Moreover, better user engagement, higher click-through rates, and conversions prompt them to use React.

Compared to other frameworks, **ReactJS assures a better performance. Why?** *Because React helps to prevent updating of DOM. So, that implies the apps will be faster and deliver better UX.* So, ReactJS appears to experience the best demand whatsoever. And if you already have a ReactJS Interview call waiting for you, then you should be reading through these **important React interview questions**.

Find the most asked React Interview Questions answered for you.

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Basic to Advanced React Interview Questions:

1. What is React?

React is an open-source front-end JavaScript library that is used for building user interfaces especially for single-page applications. It is used for handling the view layer for web and mobile apps. React was



created by [Jordan Walke](#), a software engineer working for Facebook. **React was first deployed on Facebook's News Feed in 2011 and on Instagram in 2012.**

2. What are the major features of React?

The major features of React are:

- *It uses Virtual DOM instead of Real DOM considering that Real DOM manipulations are expensive.*
- *Furthermore, React supports server-side rendering.*
- *Also, it follows unidirectional data flow or data binding.*
- *React uses reusable/composable UI components to develop the view.*

If you're someone who wants to learn about the skills, here are the [6 Essential Prerequisites For Learning ReactJS](#).

Before diving into the next question, ensure you're solid on full-stack development essentials like front-end frameworks, back-end technologies, and database management. If you are looking for a detailed Full Stack Development career program, you can join GUVI's [Full Stack Development Course](#) with placement assistance. You will be able to master the MERN stack (MongoDB, Express.js, React, Node.js) and build real-life projects.

Additionally, if you want to explore ReactJS through a self-paced course, try GUVI's self-paced [ReactJS certification course](#).

3. What is JSX?

JSX is a XML-like syntax extension to ECMAScript (the acronym stands for JavaScript XML). Basically, it just provides syntactic sugar for the `React.createElement()` function, giving us expressiveness of JavaScript along with HTML like template syntax.

In the example below text inside `<h1>` tag is returned as JavaScript function to the render function.



```
class App extends React.Component {
  render() {
    return(
      <div>
        <h1>{'Welcome to React world!'}</h1>
      </div>
    )
  }
}
```

4. What is state in React?

The **state** of a component is an object that holds some information that may change over the lifetime of the component. So, we should always try to make our state as simple as possible. In addition try to minimize the number of stateful components.

Let's create a user component with a message state:



The state is similar to props, but it is private and fully controlled by the component. i.e., It is not accessible to any component other than the one that owns and sets it.

[useState\(\) Hook in React for Beginners | React Hooks 2024](#)

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5. What are props in React?

Props are inputs to components. Moreover, they are single values or objects containing a set of values that are passed to components on creation using a naming convention similar to HTML-tag attributes. Moreover, they are data passed down from a parent component to a child component. To be precise, the primary purpose of props in React is to provide the following component functionality:



Pass custom data to your component.

- I. Trigger state changes.
- II. And use via this.props.reactProp inside component's render() method.
- III. For example, let us create an element with reactProp property:

```
<Element reactProp={"1"} />
```

This reactProp (or whatever you came up with) name then becomes a property attached to React's native props object which originally already exists on all components created using React library.props.reactProp.

6. What is the difference between state and props?

Both props and state are plain JavaScript objects. While both of them hold information that influences the output of render, they are different in their functionality concerning components. Moreover, you can pass the Props to the component similar to function parameters. Whereas you can manage the state within the component similar to variables declared within a function.

7. What is the purpose of callback function as an argument of setState()?

The callback function is invoked when setState is finished. And the component gets rendered. Since [setState\(\)](#) is asynchronous, you can use the callback function for any post action.

Note: We recommend you to use the lifecycle method rather than this callback function.

```
setState({ name: "John" }, () =>
  console.log("The name has updated and component re-rendered")
);
```

8. What is the difference between [HTML](#) and React event handling?

Below are some of the main differences between HTML and React event handling:

Here, in HTML, you can represent the event name in lowercase as a convention:



```
<button onclick='activateLasers() '>
```

Whereas, in React, it follows camelCase convention:

```
<button onClick={activateLasers}>
```

In HTML, you can return false to prevent default behavior:

```
<a href="#" onclick='console.log("The link was clicked."); r  
eturn false;' />
```

Whereas in React you must call preventDefault() explicitly:

```
function handleClick(event) {  
  event.preventDefault();  
  console.log("The link was clicked.");  
}
```

In HTML, you need to invoke the function by appending () Whereas in react you should not append() with the function name. (refer to “activateLasers” function in the first point for example)

9. What are synthetic events in React?

SyntheticEvent is a cross-browser wrapper around the browser’s native event. Its API is the same as the browser’s native event, including stopPropagation() and preventDefault(), except the events work identically across all browsers.

10. What are inline conditional expressions?

You can use either if statements or ternary expressions which are available from JS to conditionally render expressions. Apart from these approaches, you can also embed any expressions in JSX by wrapping them in curly braces and then followed by JS logical operator &&.

```
<h1>Hello!</h1>  
{  
  messages.length > 0 && !isLogin ? (  
    <h2>You have {messages.length} unread messages.</h2>  
  ) : (  
    <h2>You don't have unread messages.</h2>  
  );  
}
```



Do read the [4 Key Differences Between == and === Operators in JavaScript](#)

11. What is the use of refs?

You can use the ref to return a reference to the element. You can avoid them in most cases. However, they can be useful when you need direct access to the **DOM** element or an instance of a component.

12. How to create refs?

There are two approaches to create refs:

1. This is a recently added approach. Refs are created using `React.createRef()` method and attached to React elements via the `ref` attribute. In order to use refs throughout the component, just assign the ref to the instance property within constructor.

```
class MyComponent extends React.Component {
  constructor(props) {
    super(props)
    this.myRef = React.createRef()
  }
  render() {
    return <div ref={this.myRef} />
  }
}
```

2. You can also use the ref callbacks approach regardless of React version. For example, you can access the search bar component's input element as follows,



```

class SearchBar extends Component {
  constructor(props) {
    super(props);
    this.txtSearch = null;
    this.state = { term: '' };
    this.setInputSearchRef = e => {
      this.txtSearch = e;
    }
  }
  onChange(event) {
    this.setState({ term: this.txtSearch.value });
  }
  render() {
    return (
      <input
        value={this.state.term}
        onChange={this.onChange.bind(this)}
        ref={this.setInputSearchRef} />
    );
  }
}

```

You can also use refs in function components using closures.

Note: You can also use inline ref callbacks even though it is not a recommended approach.

13. What is the difference between Shadow DOM and Virtual DOM?

The **Shadow DOM is a browser technology**. It is specifically designed for scoping variables and CSS in web components. Thus, the Virtual DOM is a concept which libraries implement in JavaScript on top of browser APIs.

14. What is React Fiber?

Fiber is the new reconciliation engine or reimplementation of the core algorithm in React v16. The goal of React Fiber is to increase its suitability for areas like animation, layout, gestures, ability to pause, abort, or reuse work. And assign priority to different types of updates; and new concurrency primitives.

15. What is the main goal of React Fiber?

The goal of React Fiber is to increase its suitability for areas like **animation, layout, and gestures**. Also, its headline feature is



incremental rendering: the ability to split rendering work into chunks and spread it out over multiple frames.

Scroll down to find more React Interview Questions:

16. What are controlled components?

A component that controls the input elements within the forms on subsequent user input is called Controlled Component, i.e., every state mutation will have an associated handler function. For example, to write all the names in uppercase letters, we use `handleChange` as below,

```
handleChange(event) {  
  this.setState({value: event.target.value.toUpperCase()})  
}
```

17. What is the difference between `createElement` and `cloneElement`?

JSX elements will be 'transpiled' (**TRANSlate comPILED**) to `React.createElement()` functions to create React elements which are going to be used for the object representation of UI. Whereas `cloneElement` is used to clone an element and pass it new props.

18. What are the advantages of React?

Below is the list of main advantages of React:

- I. Increases the application's performance with Virtual DOM.*
- II. Also, JSX makes code easy to read and write.*
- III. It renders both on the client and server-side (SSR).*
- IV. Easy to integrate with frameworks (Angular, Backbone) since it is only a view library.*
- V. Easy to write unit and integration tests with tools such as Jest*

19. What are the limitations of React?

Apart from the advantages, there are a few limitations of React too,

- I. React is just a view library, not a full framework.*
- II. There is a learning curve for beginners who are new to web*



development.

III. Integrating React into a traditional MVC framework requires some additional configuration.

IV. The code complexity increases with inline templating and JSX.

V. Also, too many smaller components leading to over-engineering or boilerplate.

20. What are the Pointer Events supported in React?

Pointer Events provide a unified way of handling all input events. In the old days, we had a mouse and respective event listeners to handle them but nowadays we have many devices which don't correlate to having a mouse, like phones with touch surfaces or pens. We need to remember that these events will only work in browsers that support the Pointer Events specification.

The following event types are now available in React DOM:

I. `onPointerDown`

II. `onPointerMove`

III. `onPointerUp`

IV. `onPointerCancel`

V. `onGotPointerCapture`

VI. `onLostPointerCapture`

VII. `onPointerEnter`

VIII. `onPointerLeave`

IX. `onPointerOver`

X. `onPointerOut`

21. What is the difference between `super()` and `super(props)` in React using ES6 classes?

When you want to access `this.props` in the `constructor()` then you should pass props to the `super()` method.

Using `super(props)`:

```
class MyComponent extends React.Component {
  constructor(props) {
    super(props);
    console.log(this.props); // { name: 'John', ... }
  }
}
```

Using `super()`:



```
class MyComponent extends React.Component {  
  constructor(props) {  
    super()  
    console.log(this.props) // undefined  
  }  
}
```

Outside constructor() both will display same value for this.props.

22. What is the difference between React and ReactDOM?

The react package contains React.createElement(), React.Component, React.Children, and other helpers related to elements and component classes. You can think of these as the isomorphic or universal helpers that you need to build components.

So, the react-dom package contains ReactDOM.render(), and in react-dom/server we have server-side rendering support with ReactDOMServer.renderToString() and ReactDOMServer.renderToStaticMarkup().

23. Why ReactDOM is separated from React?

The React team worked on extracting all DOM-related features into a separate library called ReactDOM. React v0.14 is the first release in which the libraries are split. By looking at some of the packages, react-native, react-art, react-canvas, and react-three, it has become clear that the beauty and essence of React have nothing to do with browsers or the DOM.

So, to build more environments that React can render to, React team planned to split the main React package into two: react and react-dom. This paves the way to writing components that can be shared between the web version of React and React Native.

More React Interview Questions:

24. What is the difference between setState() and replaceState() methods?

When you use setState() the current and previous states are merged. replaceState() throws out the current state and replaces it with only what you provide. Usually, setState() is used unless you really need to



remove all previous keys for some reason. You can also set state to false/null in `setState()` instead of using `replaceState()`.

Furthermore, we have more interesting React interview questions coming up for you. So, scroll down and find out.

25. How to use https instead of http in create-react-app?

You just need to use `HTTPS=true` configuration. You can edit your `package.json` scripts section:

```
"scripts": {  
  "start": "set HTTPS=true && react-scripts start"  
}
```

or just run `set HTTPS=true && npm start`.

26. How to update a component every second?

You need to use `setInterval()` to trigger the change, but you also need to clear the timer when the component unmounts to prevent errors and memory leaks.

```
componentDidMount() {  
  this.interval = setInterval(() => this.setState({ time: Date.now() }), 1000)  
}  
  
componentWillUnmount() {  
  clearInterval(this.interval)  
}
```

27. How do you apply vendor prefixes to inline styles in React?

React does not apply vendor prefixes automatically. You need to add vendor prefixes manually.

```
<div style={{  
  transform: 'rotate(90deg)',  
  WebkitTransform: 'rotate(90deg)', // note the capital 'W'  
  msTransform: 'rotate(90deg)' // 'ms' is the only lowercase  
  vendor prefix  
}} />
```



28. Why is a component constructor called only once?

React's reconciliation algorithm assumes that without any information to the contrary, if a custom component appears in the same place on subsequent renders, it's the same component as before, so reuses the previous instance rather than creating a new one.

29. How to define constants in React?

You can use ES7 static field to define constant.

```
class MyComponent extends React.Component {  
  static DEFAULT_PAGINATION = 10  
}
```

Static fields are part of the Class Fields stage 3 proposal.

30. How to programmatically trigger click event in React?

You could use the ref prop to acquire a reference to the underlying HTMLInputElement object through a callback, store the reference as a class property, then use that reference to later trigger a click from your event handlers using the HTMLElement.click method.

This can be done in two steps:

```
<input ref={input => this.inputElement = input} />
```

Create ref in render method:

```
this.inputElement.click()
```

31. Is it possible to use async/await in plain React?

If you want to use async/await in React, then you will need Babel and transform-async-to-generator plugin. So, React Native ships with Babel and a set of transforms.

Next in our React Interview Questions:

32. What are the popular packages for animation?



React Transition Group and **React Motion** are popular animation packages in React ecosystem.

33. What is React Router?– one of the top React interview questions

React Router is a powerful routing library built on top of React that helps you add new screens and flows to your application incredibly quickly, all while keeping the URL in sync with what's being displayed on the page.

34. How React Router is different from history library?

[React Router](#) is a wrapper around the history library which handles interaction with the browser's window.history with its browser and hash histories. It also provides memory history which is useful for environments that don't have global history, such as mobile app development (React Native) and unit testing with Node.

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For more React Interview Questions

That's not all! We have a few more React Interview questions answered with us. We will be bringing it in our next part of this Most Important React Interview Questions series – [Most Important React Interview Questions Of 2024! \[Part-2\]](#)

Have a query? Why don't you drop it with us in the comments below. You can also take Expert advice from our Tech Geeks– just



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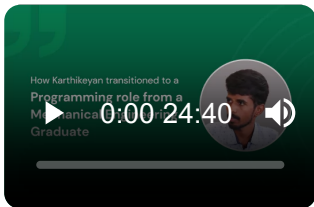
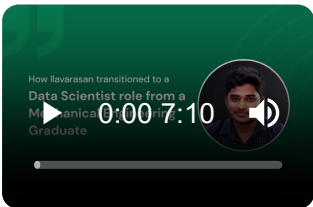
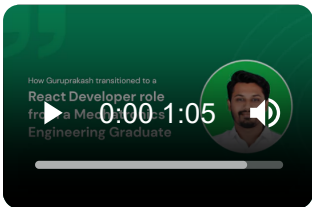
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A traveler, and explorer, Archana is an active writer at GUVI. You can usually find her with a book/kindle binge-reading (with an affinity to the mystery/humor).

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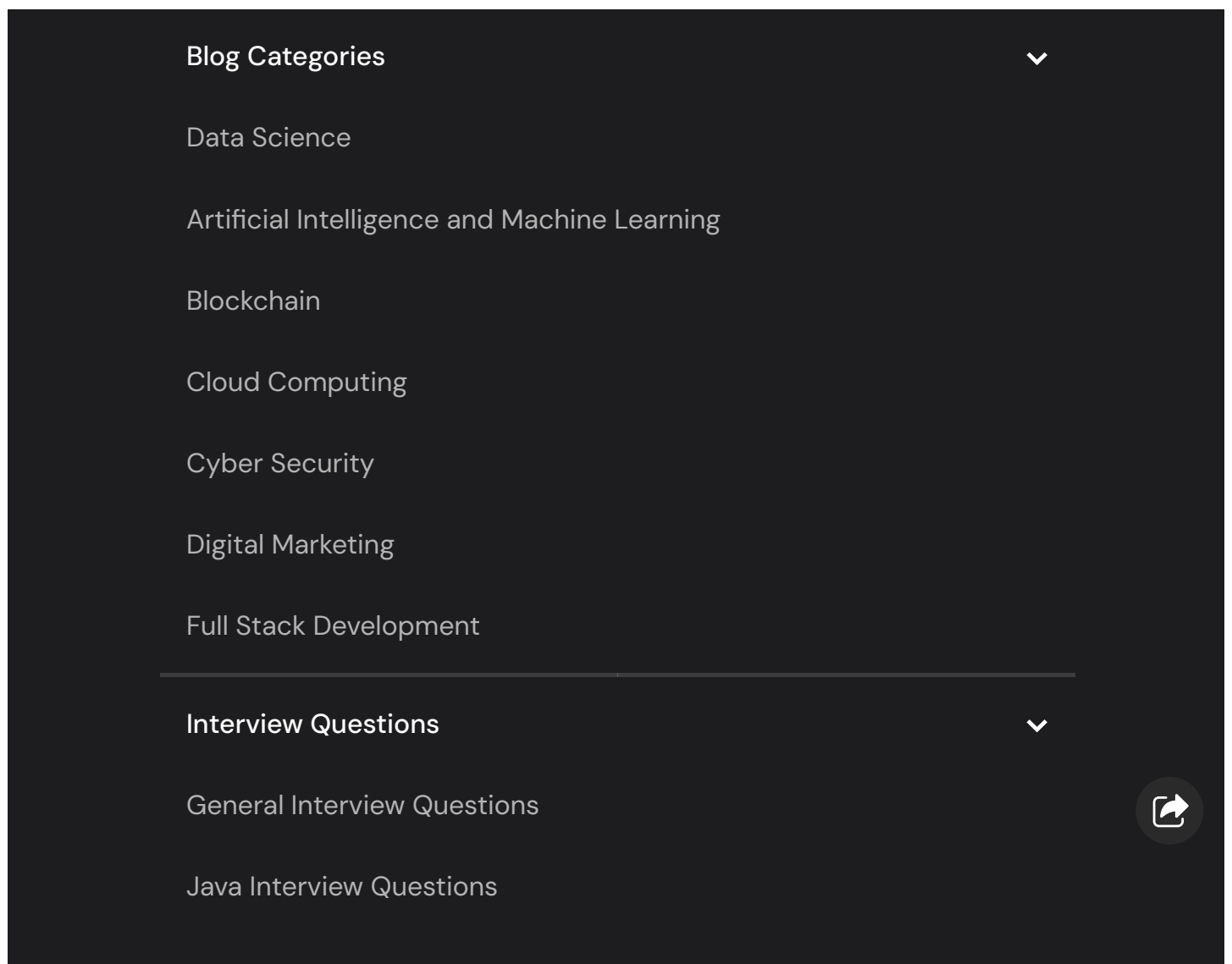
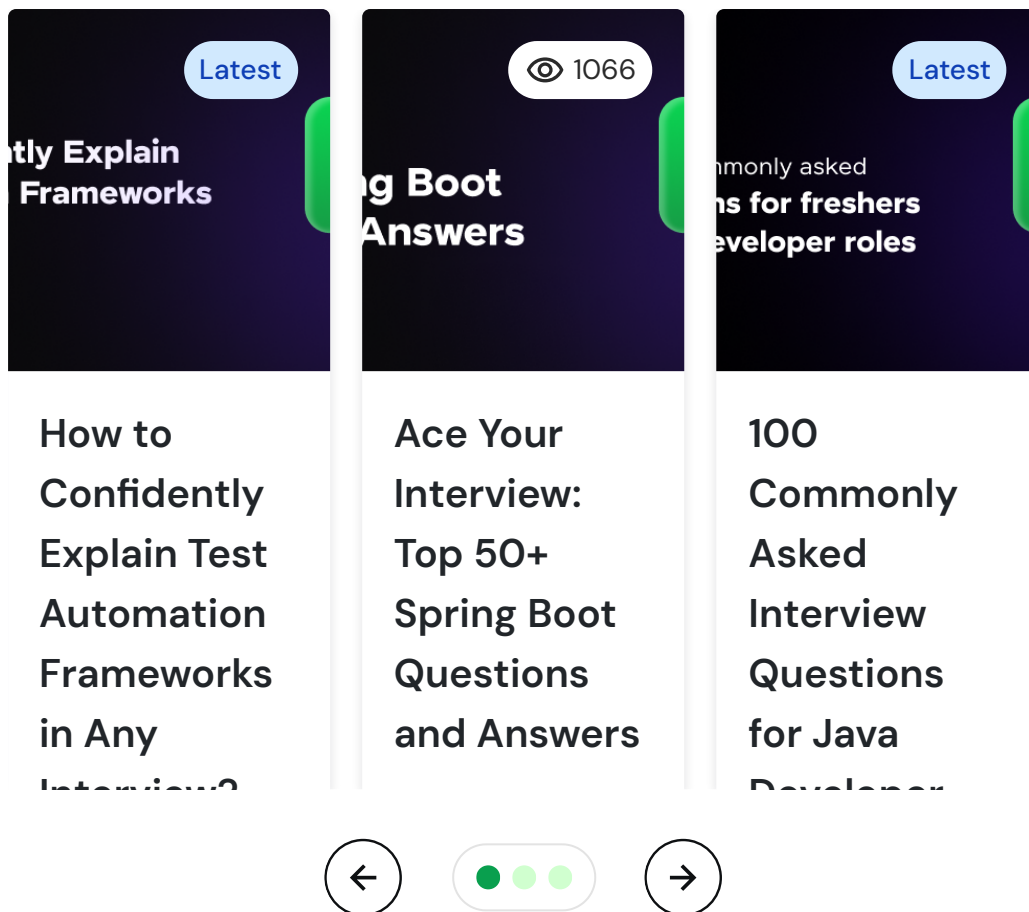


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